DIABETES PREDICTION USING MACHINE LEARNING

A College AIML Project

Overview and Objectives

Diabetes is a chronic metabolic disorder affecting millions worldwide. Early and accurate prediction of diabetes is crucial for timely intervention and management. This project aims to develop a machine learning model capable of predicting diabetes using various health-related features.

The objectives of this project are to:

- Utilize machine learning techniques for diabetes prediction.
- Preprocess and analyze a diabetes dataset.
- Build and evaluate different classification models.
- Identify key features contributing to diabetes.
- Provide insights for early diabetes detection.

Dataset Description

The dataset used in this project contains various features related to individuals' health. These features include:

- **Pregnancies:** Number of times pregnant.
- **Glucose:** Plasma glucose concentration after a 2-hour oral glucose tolerance test.
- **BloodPressure:** Diastolic blood pressure (mm Hg).
- **SkinThickness:** Triceps skin fold thickness (mm).
- **Insulin:** 2-Hour serum insulin (mu U/ml).
- **BMI:** Body mass index (weight in kg/(height in m)^2).
- **DiabetesPedigreeFunction:** Diabetes pedigree function (a function which scores likelihood of diabetes based on family history).
- **Age:** Age in years.
- **Outcome:** Class variable (0 or 1), where 1 indicates diabetes.

Each feature plays a significant role in determining the likelihood of diabetes. For instance, high glucose levels, elevated BMI, and a family history of diabetes are strong indicators.

Data Preprocessing and Model Building

Data Preprocessing Steps:

- Handling Missing Values
- Feature Scaling
- Data Splitting

The data preprocessing stage involves cleaning and preparing the data for model training. Missing values were imputed using appropriate methods (e.g., mean or median imputation). Feature scaling, using StandardScaler, was applied to standardize the range of independent variables. Finally, the dataset was split into training and testing sets to evaluate the model's performance on unseen data.

Model Building:

Various classification algorithms were employed to build the diabetes prediction model. These included:

- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)

Each model was trained on the preprocessed training data and evaluated on the testing data using metrics such as accuracy, precision, recall, and F1-score. Hyperparameter tuning was performed using techniques like GridSearchCV to optimize model performance.

Results and Conclusion

The machine learning model successfully predicts whether a person is diabetic or not based on the input features. Different algorithms yielded varying levels of performance, with Random Forest and SVM showing promising results. The accuracy and other evaluation metrics demonstrate the model's potential for assisting in early diabetes detection and management.

Further improvements can be achieved by exploring additional features, advanced preprocessing techniques, and ensemble methods.

Technologies Used

The following technologies were used in this project:

- **Programming Language:** Python 3
- **Platform:** Google Colab
- **Libraries:**
  - NumPy: For numerical computations.
  - Pandas: For data manipulation and analysis.

- Scikit-learn: For machine learning algorithms and model evaluation.
- Matplotlib & Seaborn: For data visualization.

Summary

This project successfully demonstrates the application of machine learning techniques for diabetes prediction. By leveraging health-related features and employing various classification algorithms, the model provides valuable insights for early diabetes detection and management. The use of Python, along with libraries like NumPy, Pandas, and Scikit-learn, facilitated efficient data preprocessing, model building, and evaluation. The results highlight the potential of machine learning in improving healthcare outcomes.